

Medical Assistive Robot

→ Human-Following Smart Dustbin.

Project title: Human-Following Smart Dustbin

Course Name: Robotics Lab

Course Code: CSE 438

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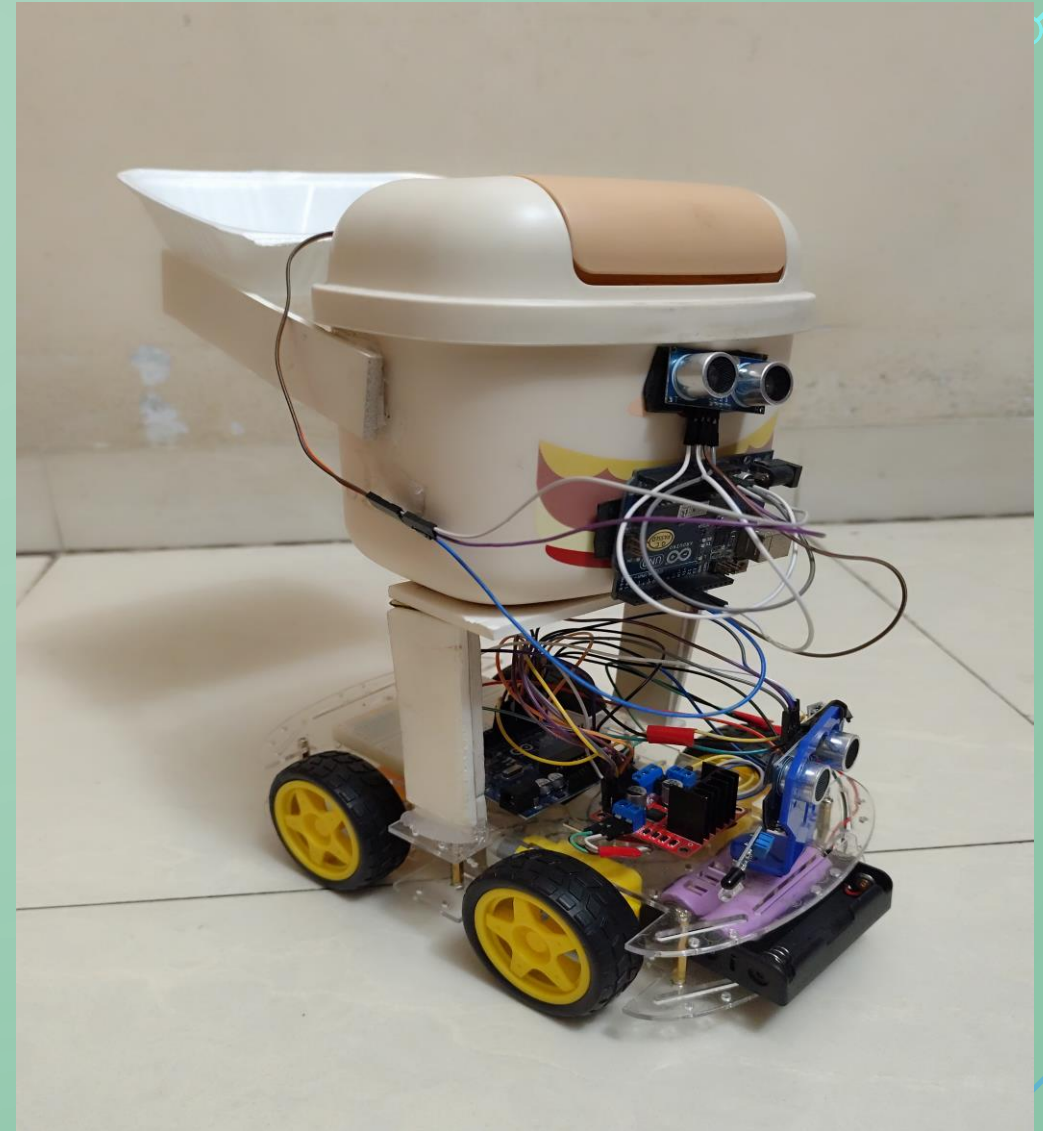
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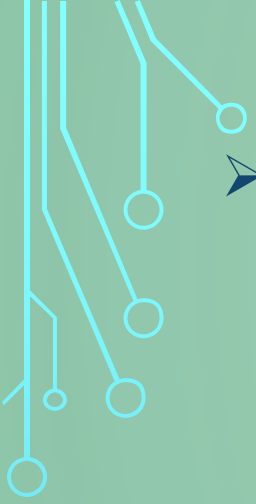
Executive Summary

This project develops a smart assistive robot that can follow a user and support hygienic waste disposal. The system reads distance and sensor values, controls robot movement, and opens or closes the dustbin lid automatically. The main goal is to reduce manual contact and make waste disposal safer and easier.



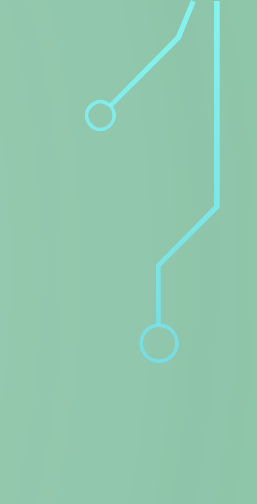
Project Goals:

- ☐ Automate medical waste collection in hospitals.
- ☐ Reduce human effort and cross-contamination risks.
- ☐ Provide fast, safe, and hygienic waste disposal.
- ☐ Help healthcare workers and patients avoid touching biohazardous waste.
- ☐ Show how a microcontroller-based robot can work in real medical settings.
- ☐ Improve patient safety through autonomous robot assistance

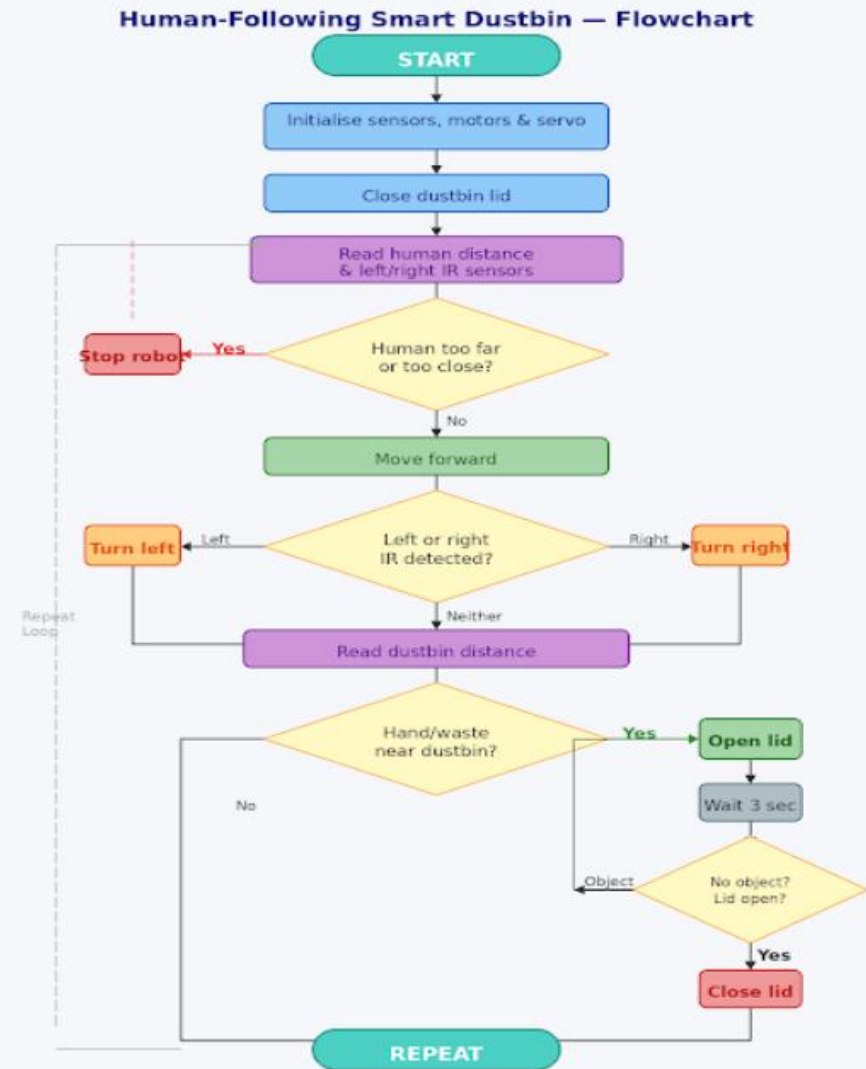
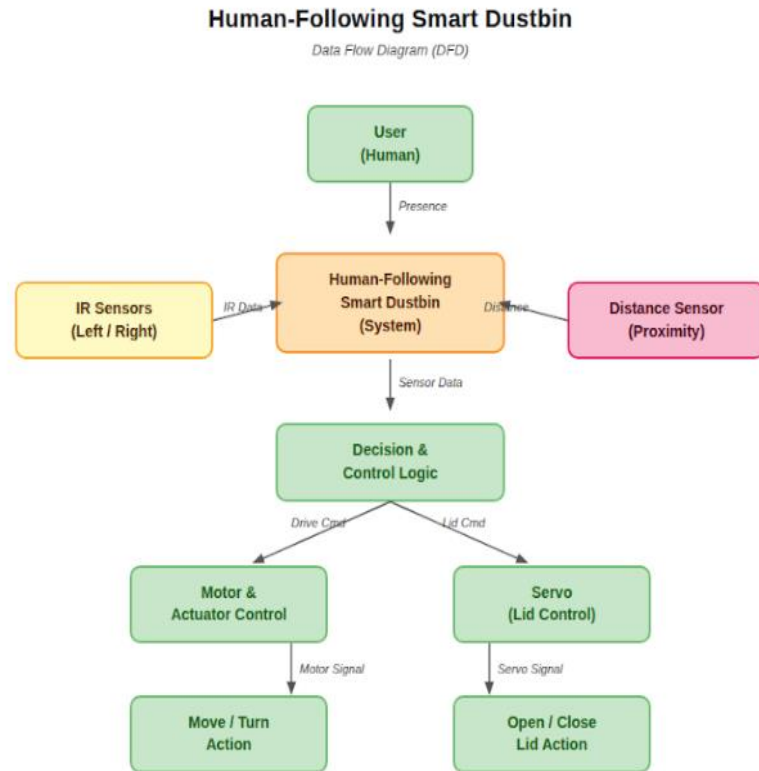


➤ **My Contribution in This Project:**

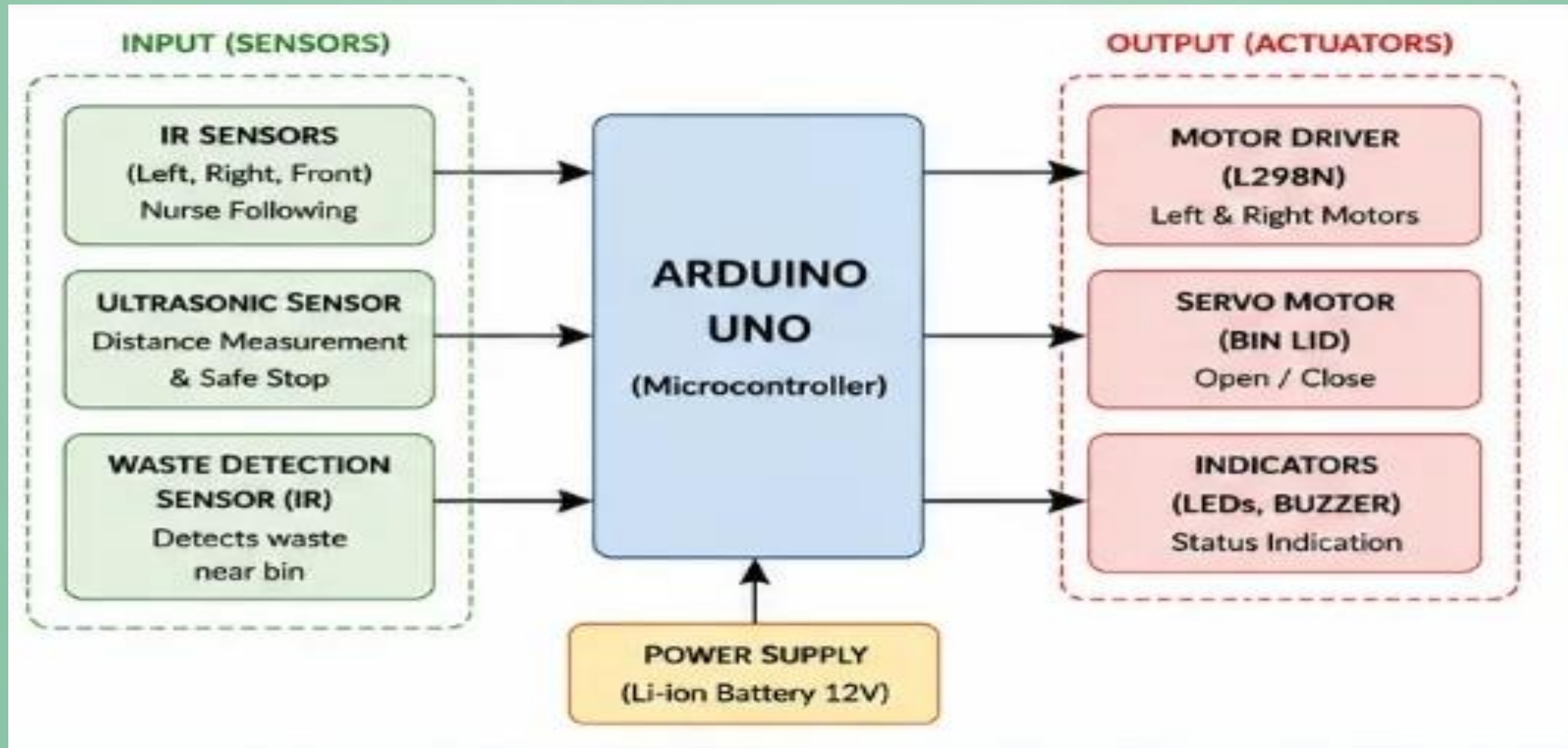
My contribution was mainly in the idea and survey part. I worked on:

- Related work analysis
 - Workflow and architecture design
 - Project planning (gantt chart)
 - Documentation
 - Future scope analysis
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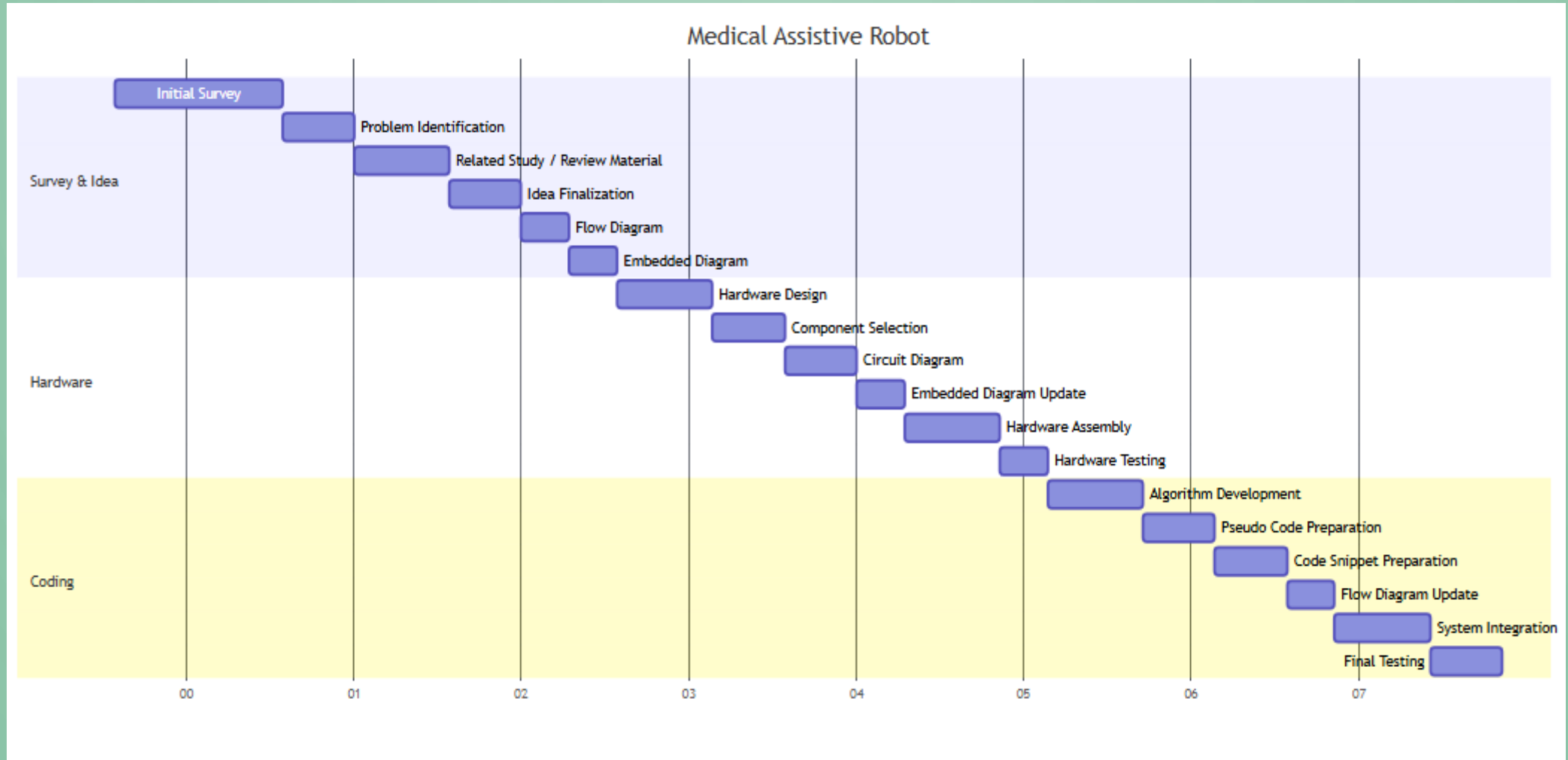
Data Flow Diagram and Flowchart



➤ Embedded Diagram

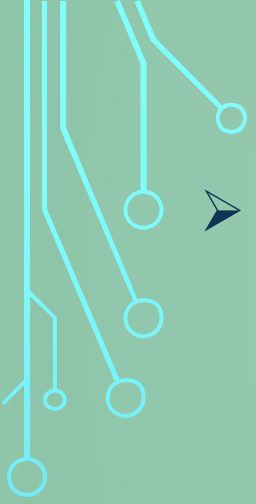


➤ Gantt Chart



➤ Future Works

<u>Future Work</u>	<u>Simple Description</u>
• Voice Control System	The robot can follow voice commands from users or hospital staff.
• Mobile App Monitoring	Users can monitor and control the robot using a smartphone app.
• Camera Monitoring System	A camera can help monitor patients and surroundings for better safety.
• GPS / Indoor Navigation	The robot can move automatically to specific rooms or hospital locations.
• IoT Connectivity	Robot data and status can be monitored through the internet in real time.
• Emergency Alert System	The robot can send alerts during emergency situations in hospitals.
• Automatic Waste Segregation	The dustbin can automatically separate medical and normal waste.



➤ Conclusion

These future improvements will make the robot smarter, more efficient, and more useful in modern hospital environments.

➤ References

- <https://youtu.be/mHaeDoFOVX0?si=cqccbFWXPTjQvu9I>
 - <https://youtu.be/yAV5aZ0unag?si=AmRvg2fXpFjX-qa8>
 - <https://youtu.be/6Nh3nOrR9EE?si=dvioz9FLGzwsGG6x>
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➤ Videos



<https://drive.google.com/file/d/1kpYValiLjYHYRoYT2essKdnej0vyY8Vy/view?usp=sharing>



https://drive.google.com/file/d/1OtVW5VENEGhgHX3zZigZDswcLK4Pq_Mu/view?usp=sharing

➤ Visual Proof



The background is a solid teal color. In the four corners, there are decorative line art elements resembling circuit boards or neural networks. These elements consist of thin, light blue lines that branch out and terminate in small circles. The lines are more densely packed in the top-left and bottom-left corners and more sparse in the top-right and bottom-right corners.

Thank you